Multiple regression

R Markdown

```
mydata<-read.csv("contents.csv")
m1<-lm(Y~., data = mydata) # Y is a function of all independent variables)
summary(m1)
```

```
##
## Call:
## lm(formula = Y ~ ., data = mydata)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.63552 -0.11390 -0.03933  0.15472  0.45138
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 20.24576    0.33020   61.31   <2e-16 ***
## X1           0.54727    0.03278   16.70   <2e-16 ***
## X2           0.70216    0.03169   22.16   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2255 on 47 degrees of freedom
## Multiple R-squared:  0.9523, Adjusted R-squared:  0.9502
## F-statistic: 468.9 on 2 and 47 DF,  p-value: < 2.2e-16
```

X1 is signifcant (p=0.000) and thus should not be considered to be dropped from the model

X2 is signifcant (p=0.000) and thus should not be considered to be dropped from the model

Check for multicollinearity

```
check_collinearity(m1) # function from the performance library
```

```
## # Check for Multicollinearity
##
## Low Correlation
##
##  Term  VIF Increased SE Tolerance
##    X1 1.04         1.02      0.97
##    X2 1.04         1.02      0.97
```

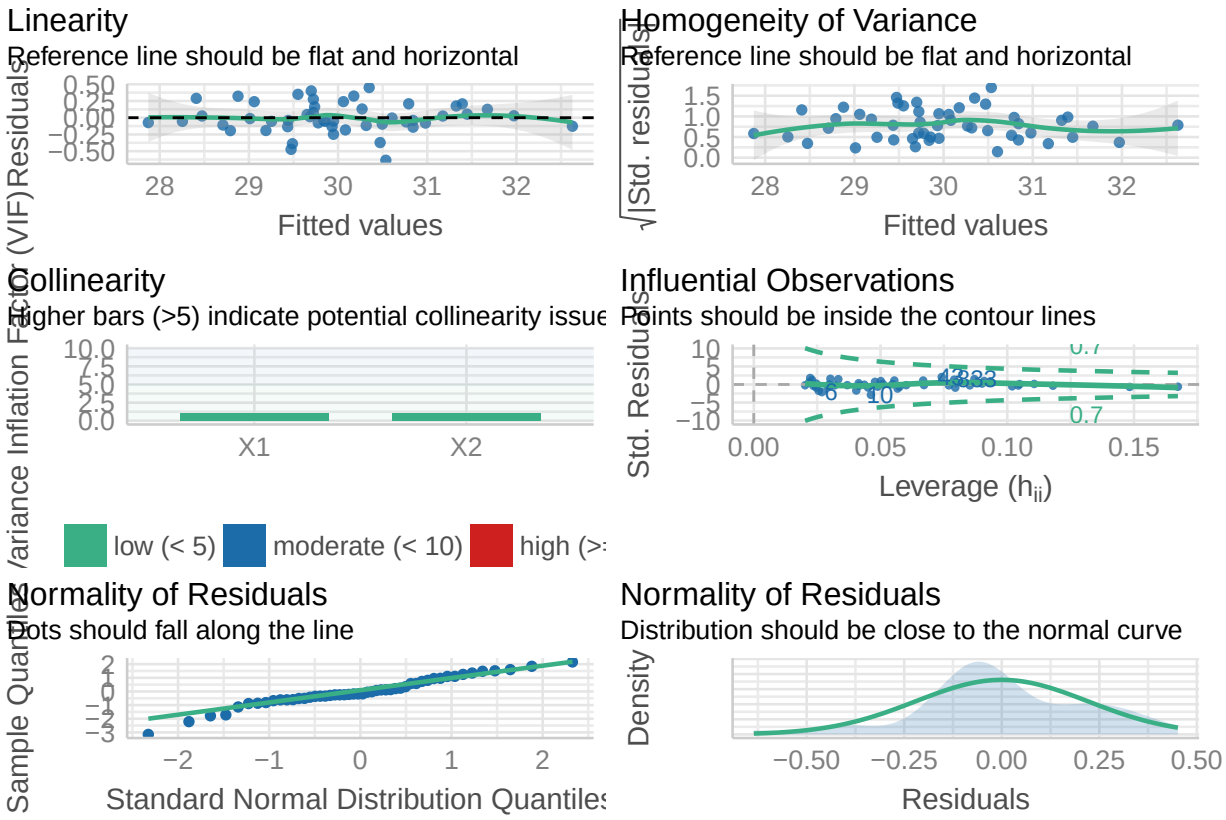
The correlation between the predictor variables is low, thus no issue of multicollinearity.

As a general rule A VIF (variance inflation factor) value greater than 5 indicates potentially severe correlation between a given predictor variable and other predictor variables in the model (VIF $>$ 5 multicollinearity is present).

If multicollinearity you could consider removing a variable giving similar information to another variable

Check other assumptions

```
check_model(m1) # function from the performance library
```



From the graphics above no major concern that the assumptions of multiple regression have been violated.